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**Le T. P. Ngoc, Nguyen A. Triet, Nguyen
T. Duy & Nguyen T. Long**

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An N -Order Iterative Scheme for a Nonlinear Love Equation

Le T. P. Ngoc¹ · Nguyen A. Triet² · Nguyen T. Duy^{3,4} ·
Nguyen T. Long⁴

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Abstract In this paper, we consider an initial and boundary value problem for a nonlinear Love equation, with the source term containing four variables $f = f(x, t, u, u_t)$. Motivated by recent results for Love equation, we establish here a high-order iterative scheme in order to obtain a convergent sequence at a rate of order N to a local weak solution of the model.

Keywords Nonlinear Love equation · Faedo–Galerkin method · The convergence of order N

Mathematics Subject Classification (2010) 35L20 · 35L70 · 35Q72

✉ Nguyen T. Long
longnt2@gmail.com

Le T. P. Ngoc
ngoc1966@gmail.com

Nguyen A. Triet
anhtriet1@gmail.com

Nguyen T. Duy
tuanduy2312@gmail.com

¹ University of Khanh Hoa, 01 Nguyen Chanh Str., Nha Trang City, Vietnam

² Department of Mathematics, University of Architecture of Ho Chi Minh City, 196 Pasteur Str., Dist. 3, Ho Chi Minh City, Vietnam

³ Department of Fundamental Sciences, University of Finance and Marketing, 306 Nguyen Trong Tuyen Str., Dist. Tan Binh, Ho Chi Minh City, Vietnam

⁴ Department of Mathematics and Computer Science, University of Natural Science, Vietnam National University Ho Chi Minh City, 227 Nguyen Van Cu Str., Dist.5, Ho Chi Minh City, Vietnam

1 Introduction

In this paper, we consider the following nonlinear Love equation with initial conditions and homogeneous Dirichlet boundary conditions

$$u_{tt} - u_{xx} - u_{xxt} = f(x, t, u, u_t), \quad 0 < x < 1, 0 < t < T, \quad (1)$$

$$u(0, t) = u(1, t) = 0, \quad (2)$$

$$u(x, 0) = \tilde{u}_0(x), \quad u_t(x, 0) = \tilde{u}_1(x), \quad (3)$$

where $\tilde{u}_0, \tilde{u}_1, f$ are given functions. Equation (1) is related to the Love equation

$$u_{tt} - \frac{E}{\rho} u_{xx} - 2\mu^2 k^2 u_{xxt} = 0,$$

presented by V. Radochová in 1978 (see [15]). Equations of Love waves or Love type waves have been studied by many authors, we refer to [1–4, 6, 10–12, 14, 16] and the references therein.

Motivated by results for Love equation in [10, 11], where recurrent sequences converge at a rate of order 1, in this paper we establish a high-order iterative scheme in order to obtain a convergent sequence at a rate of order N to a local weak solution of Problem (1)–(3). This scheme is established based on a high-order method for solving the operator equation $F(x) = 0$, it also has been applied in [9, 17, 18] and some other works. It is well known that Newton's method and its variants are used to solve nonlinear operator equations, see [13] and the references therein. In the case $\lim_{n \rightarrow \infty} u_n = u$, one speaks of *convergence of order* N if $|u_{n+1} - u| \leq C|u_n - u|^N$ for some $C > 0$ and all large N . In the special cases $N = 1$ with $C < 1$ and $N = 2$, one also speaks of linear and quadratic convergence, respectively, see [5]. Here, we shall associate to (1) a recurrent sequence $\{u_m\}$ defined by

$$\begin{aligned} & \frac{\partial^2 u_m}{\partial t^2} - \frac{\partial^2 u_m}{\partial x^2} - \frac{\partial^4 u_m}{\partial t^2 \partial x^2} \\ &= \sum_{i+j \leq N-1} \frac{1}{i!j!} \frac{\partial^{i+j} f}{\partial u^i \partial u_t^j} \left(x, t, u_{m-1}, \frac{\partial u_{m-1}}{\partial t} \right) (u_m - u_{m-1})^i \left(\frac{\partial u_m}{\partial t} - \frac{\partial u_{m-1}}{\partial t} \right)^j, \quad (4) \end{aligned}$$

$0 < x < 1, 0 < t < T$, with u_m satisfying (2), (3), and $u_0 \equiv 0$. If $f \in C^N([0, 1] \times \mathbb{R}_+ \times \mathbb{R}^2)$, we prove that the sequence $\{u_m\}$ converges at a rate of order N to a local weak solution of Problem (1)–(3). The main result is given in Theorems 1 and 2. We note more that the result obtained here is local (in time T small enough), because T is chosen corresponding to the size of the initial data, see (19) and (35) in Section 2. In our proofs, the Faedo–Galerkin approximation method associated to a priori estimates, weak convergence, compactness techniques, and a known fixed point theorem are used. Furthermore, by the term $\frac{\partial^4 u_m}{\partial t^2 \partial x^2}$ appearing in (4), we combine the arguments in [9, 17, 18] and some necessary techniques in the estimates, see Lemma 3 below.

2 A High-Order Iterative Scheme

First, we put $\Omega = (0, 1)$ and denote the usual function spaces used in this paper by the notations $L^p = L^p(\Omega)$, $H^m = H^m(\Omega)$. Let $\langle \cdot, \cdot \rangle$ be either the scalar product in L^2 or the dual pairing of a continuous linear functional and an element of a function space. The notation $\| \cdot \|$ stands for the norm in L^2 , $\| \cdot \|_X$ is the norm in the Banach space X , and X' is

the dual space of X . We denote by $L^p(0, T; X)$, $1 \leq p \leq \infty$ for the Banach space of real functions $u : (0, T) \rightarrow X$ measurable, such that

$$\|u\|_{L^p(0,T;X)} = \left(\int_0^T \|u(t)\|_X^p dt \right)^{1/p} < \infty \quad \text{for } 1 \leq p < \infty,$$

and

$$\|u\|_{L^\infty(0,T;X)} = \operatorname{ess\,sup}_{0 < t < T} \|u(t)\|_X \quad \text{for } p = \infty.$$

With $f \in C^k([0, 1] \times \mathbb{R}_+ \times \mathbb{R}^2)$, $f = f(x, t, u, v)$, we put $D_1 f = \frac{\partial f}{\partial x}$, $D_2 f = \frac{\partial f}{\partial t}$, $D_3 f = \frac{\partial f}{\partial u}$, $D_4 f = \frac{\partial f}{\partial v}$ and $D^\alpha f = D_1^{\alpha_1} \dots D_4^{\alpha_4} f$; $\alpha = (\alpha_1, \dots, \alpha_4) \in \mathbb{Z}^4$, $|\alpha| = \alpha_1 + \dots + \alpha_4 = k$, $D^{(0,0,0,0)} f = f$.

On H^1 , the following norm is used

$$\|v\|_{H^1} = \left(\|v\|^2 + \|v_x\|^2 \right)^{1/2}.$$

Note that, on $H_0^1 = \{v \in H^1 : v(0) = v(1) = 0\}$, two norms $v \mapsto \|v\|_{H^1}$ and $v \mapsto \|v_x\|$ are equivalent. We have the following lemmas, their proof can be found in [7].

Lemma 1 *The imbedding $H^1 \hookrightarrow C^0([0, 1])$ is compact. Furthermore,*

- (i) $\|v\|_{C^0([0,1])} \leq \sqrt{2}\|v\|_{H^1}$ for all $v \in H^1$,
- (ii) $\|v\|_{C^0([0,1])} \leq \|v_x\|$ for all $v \in H_0^1$.

Lemma 2 *For all $v \in H_0^1 \cap H^2$, the following inequalities hold*

- (i) $\|v_x\| \leq \|v_{xx}\|$,
- (ii) $\|v_x\|_{C^0([0,1])} \leq 2\|v_{xx}\|$.

Next, we consider $T^* > 0$ fixed. For each $M > 0$ given, we define

$$K_0(M, f) = \sup_{(x,t,u,v) \in [0,1] \times [0,T^*] \times [-M,M]^2} |f(x, t, u, v)|,$$

$$K_M(f) = \sum_{|\alpha| \leq N} K_0(M, D^\alpha f),$$

$$W(M, T) = \{v \in L^\infty(0, T; H_0^1 \cap H^2) : v_t \in L^\infty(0, T; H_0^1 \cap H^2), v_{tt} \in L^\infty(0, T; H_0^1), \\ \text{with } \|v\|_{L^\infty(0,T;H_0^1 \cap H^2)}, \|v_t\|_{L^\infty(0,T;H_0^1 \cap H^2)}, \|v_{tt}\|_{L^\infty(0,T;H_0^1)} \leq M\},$$

$$W_1(M, T) = \{v \in W(M, T) : v_{tt} \in L^\infty(0, T; H_0^1 \cap H^2)\}$$

for every $T \in (0, T^*]$.

And now, we present the main result in this section.

Theorem 1 *Suppose that $\tilde{u}_0, \tilde{u}_1 \in H_0^1 \cap H^2$ and $f \in C^N([0, 1] \times \mathbb{R}_+ \times \mathbb{R}^2)$ satisfying $f(0, t, 0, 0) = f(1, t, 0, 0) = 0 \forall t \geq 0$.*

Then, there are positive constants M, T such that there exists a sequence $\{u_m\} \subset W_1(M, T)$ defined as follows

- (i) *The first term $u_0 = 0$;*
- (ii) *With the given term*

$$u_{m-1} \in W_1(M, T), \tag{5}$$

there exists the next term $u_m \in W_1(M, T)$ ($m \geq 1$) satisfying

$$\begin{cases} \langle u_m''(t), w \rangle + \langle u_{mx}(t) + u_{mx}''(t), w_x \rangle = \langle F_m(t), w \rangle & \forall w \in H_0^1, \\ u_m(0) = \tilde{u}_0, \quad u_m'(0) = \tilde{u}_1, \end{cases} \quad (6)$$

in which

$$F_m(x, t) = \sum_{i+j \leq N-1} \frac{1}{i!j!} D_3^i D_4^j f[u_{m-1}](u_m - u_{m-1})^i (u_m' - u_{m-1}')^j, \quad (7)$$

where $f[u] = f(x, t, u, u_t)$, $D^\alpha f[u] = D_1^{\alpha_1} \dots D_4^{\alpha_4} f[u] = D^\alpha f(x, t, u, u_t)$.

Proof Approximating solutions. Let $\{w_j\}$ be a basis of H_0^1 , formed by eigenfunctions w_j of the operator $-\Delta = -\frac{\partial^2}{\partial x^2} : -\Delta w_j = \lambda_j w_j$, $w_j \in H_0^1 \cap H^2$, $w_j(x) = \sqrt{2} \sin(j\pi x)$, $\lambda_j = (j\pi)^2$, $j = 1, 2, 3, \dots$

If

$$u_m^{(k)}(t) = \sum_{j=1}^k c_{mj}^{(k)}(t) w_j$$

satisfies the following system

$$\begin{cases} \langle \ddot{u}_m^{(k)}(t), w_j \rangle + \langle u_{mx}^{(k)}(t) + \ddot{u}_{mx}^{(k)}(t), w_{jx} \rangle = \langle F_m^{(k)}(t), w_j \rangle, \\ u_m^{(k)}(0) = \tilde{u}_{0k}, \quad \dot{u}_m^{(k)}(0) = \tilde{u}_{1k}, \quad j = 1, 2, \dots, k, \end{cases} \quad (8)$$

where

$$\begin{cases} \tilde{u}_{0k} = \sum_{j=1}^k \alpha_j^{(k)} w_j \longrightarrow \tilde{u}_0 \text{ strongly } H_0^1 \cap H^2, \\ \tilde{u}_{1k} = \sum_{j=1}^k \beta_j^{(k)} w_j \longrightarrow \tilde{u}_1 \text{ strongly } H_0^1 \cap H^2, \end{cases} \quad (9)$$

and

$$F_m^{(k)}(x, t) = \sum_{i+j \leq N-1} D^{ij} f[u_{m-1}] \left(u_m^{(k)} - u_{m-1} \right)^i \left(\dot{u}_m^{(k)} - u_{m-1}' \right)^j, \quad (10)$$

with $D^{ij} f = \frac{1}{i!j!} D_3^i D_4^j f = \frac{1}{i!j!} \frac{\partial^{i+j} f}{\partial u^i \partial u_t^j}$, $i + j \leq N$, $D^{00} f = f$, then $c_{mj}^{(k)}$ satisfies the following system

$$\begin{cases} \ddot{c}_{mj}^{(k)}(t) + \mu_j^2 c_{mj}^{(k)}(t) = f_{mj}^{(k)}(t), \\ c_{mj}^{(k)}(0) = \alpha_j^{(k)}, \quad \dot{c}_{mj}^{(k)}(0) = \beta_j^{(k)}, \quad 1 \leq j \leq k, \end{cases} \quad (11)$$

where

$$f_{mj}^{(k)}(t) = \frac{1}{1 + \lambda_j} \langle F_m^{(k)}(t), w_j \rangle, \quad \mu_j^2 = \frac{\lambda_j}{1 + \lambda_j}, \quad \lambda_j = (j\pi)^2, \quad 1 \leq j \leq k.$$

System (11) is equivalent to the system of integral equations

$$c_{mj}^{(k)}(t) = \alpha_j^{(k)} \cos(\mu_j t) + \beta_j^{(k)} \frac{\sin(\mu_j t)}{\mu_j} + \int_0^t \frac{\sin(\mu_j(t-s))}{\mu_j} f_{mj}^{(k)}(s) ds, \quad 1 \leq j \leq k, \quad (12)$$

or

$$c = F[c],$$

in which

$$\begin{cases} F[c] = (F_1[c], \dots, F_k[c]), \\ c = (c_{m1}^{(k)}, \dots, c_{mk}^{(k)}) \equiv (c_1, \dots, c_k), \\ \alpha = (\alpha_1^{(k)}, \dots, \alpha_k^{(k)}) \equiv (\alpha_1, \dots, \alpha_k), \\ \beta = (\beta_1^{(k)}, \dots, \beta_k^{(k)}) \equiv (\beta_1, \dots, \beta_k), \end{cases}$$

and

$$\begin{cases} F_j[c](t) = q_j(t) + \frac{1}{1+\lambda_j} \int_0^t \frac{\sin(\mu_j(t-s))}{\mu_j} \langle F[c, c'](s), w_j \rangle ds, \quad 1 \leq j \leq k, \\ q_j(t) = \alpha_j \cos(\mu_j t) + \beta_j \frac{\sin(\mu_j t)}{\mu_j}, \quad 1 \leq j \leq k, \\ F[c, c'](x, t) = \sum_{i+j \leq N-1} D^{ij} f[u_{m-1}](u(t) - u_{m-1})^i (u'(t) - u'_{m-1})^j, \\ u(t) = \sum_{j=1}^k c_j(t) w_j. \end{cases}$$

Applying the known contraction principle, we imply that the system (12) has a unique solution $c_{mj}^{(k)}(t)$ in $[0, T_m^{(k)}]$, with certain $T_m^{(k)} \in (0, T]$.

Indeed, for every $T_m^{(k)} \in (0, T]$ and $\rho > 0$ chosen later, we set

$$X = C^1([0, T_m^{(k)}]; \mathbb{R}^k), \quad S = \{c \in X : \|c\|_X \leq \rho\},$$

with

$$\|c\|_X = \sup_{0 \leq t \leq T_m^{(k)}} (|c(t)|_1 + |c'(t)|_1), \quad |c(t)|_1 = \sum_{j=1}^k |c_j(t)|, \quad |c'(t)|_1 = \sum_{j=1}^k |c'_j(t)|.$$

It is clear that S is a closed nonempty subset of X and $F : X \rightarrow X$. We can choose $\rho > 0$ and $T_m^{(k)} > 0$ such that $F : S \rightarrow S$ is contractive. Then, system (12) has a unique solution $c_{mj}^{(k)}(t)$ in $[0, T_m^{(k)}]$. We deduce that system (8) has a unique solution $u_m^{(k)}(t)$ in $[0, T_m^{(k)}]$.

The following estimates allow one to take $T_m^{(k)} = T$ independent of m and k .

A priori estimates. Multiplying (8)₁ by $\dot{c}_m^{(k)}(t)$, summing up for $j = 1, \dots, k$ and then integrating with respect to the time variable from 0 to t , it gives

$$p_m^{(k)}(t) \equiv \|\dot{u}_m^{(k)}(t)\|^2 + \|u_{mx}^{(k)}(t)\|^2 + \|\dot{u}_{mx}^{(k)}(t)\|^2 = p_m^{(k)}(0) + 2 \int_0^t \langle F_m^{(k)}(s), \dot{u}_m^{(k)}(s) \rangle ds. \quad (13)$$

By replacing w_j in (8)₁ by $-w_{jxx}$, we obtain

$$\langle \ddot{u}_{mx}^{(k)}(t), w_{jx} \rangle + \langle u_{mxx}^{(k)}(t) + \ddot{u}_{mxx}^{(k)}(t), w_{jxx} \rangle = \langle F_{mx}^{(k)}(t), w_{jx} \rangle, \quad 1 \leq j \leq k,$$

and then, with the same way as (8)₁, it leads to

$$q_m^{(k)}(t) \equiv \|\dot{u}_{mx}^{(k)}(t)\|^2 + \|u_{mxx}^{(k)}(t)\|^2 + \|\dot{u}_{mxx}^{(k)}(t)\|^2 = q_m^{(k)}(0) + 2 \int_0^t \langle F_{mx}^{(k)}(s), \dot{u}_{mx}^{(k)}(s) \rangle ds. \quad (14)$$

Now differentiating (8)₁ with respect to t , we have

$$\langle \ddot{u}_m^{(k)}(t), w_j \rangle + \langle \dot{u}_{mx}^{(k)}(t) + \ddot{u}_{mx}^{(k)}(t), w_{jx} \rangle = \langle \dot{F}_m^{(k)}(t), w_j \rangle \quad (15)$$

for all $1 \leq j \leq m$. Also multiplying (15) by $\ddot{c}_{mj}^{(k)}(t)$ and summing up for $j = 1, \dots, k$, and integrating from 0 to t , it implies that

$$r_m^{(k)}(t) \equiv \|\ddot{u}_m^{(k)}(t)\|^2 + \|\dot{u}_{mx}^{(k)}(t)\|^2 + \|\ddot{u}_{mx}^{(k)}(t)\|^2 = r_m^{(k)}(0) + 2 \int_0^t \langle \dot{F}_{mx}^{(k)}(s), \ddot{u}_m^{(k)}(s) \rangle ds. \quad (16)$$

Combining (13), (14), and (16), we find

$$\begin{aligned}
 S_m^{(k)}(t) &\equiv p_m^{(k)}(t) + q_m^{(k)}(t) + r_m^{(k)}(t) \\
 &= S_m^{(k)}(0) + 2 \int_0^t \left\langle F_m^{(k)}(s), \dot{u}_m^{(k)}(s) \right\rangle ds \\
 &\quad + 2 \int_0^t \left\langle F_{mx}^{(k)}(s), \dot{u}_{mx}^{(k)}(s) \right\rangle ds + 2 \int_0^t \left\langle \dot{F}_m^{(k)}(s), \ddot{u}_m^{(k)}(s) \right\rangle ds \\
 &\equiv S_m^{(k)}(0) + \sum_{j=1}^3 I_j.
 \end{aligned} \tag{17}$$

We first estimate $\xi_m^{(k)} = \|\ddot{u}_m^{(k)}(0)\|^2 + \|\ddot{u}_{mx}^{(k)}(0)\|^2$.

Letting $t \rightarrow 0_+$ in (8)₁, multiplying the result by $\ddot{c}_{mj}^{(k)}(0)$, we get

$$\left\| \ddot{u}_m^{(k)}(0) \right\|^2 + \left\| \ddot{u}_{mx}^{(k)}(0) \right\|^2 + \left\langle u_{mx}^{(k)}(0), \ddot{u}_{mx}^{(k)}(0) \right\rangle = \left\langle F_m^{(k)}(0), \ddot{u}_{mx}^{(k)}(0) \right\rangle.$$

This implies that

$$\begin{aligned}
 \xi_m^{(k)} &= \left\| \ddot{u}_m^{(k)}(0) \right\|^2 + \left\| \ddot{u}_{mx}^{(k)}(0) \right\|^2 \leq \left(\left\| u_{mx}^{(k)}(0) \right\| + \left\| F_m^{(k)}(0) \right\| \right) \left\| \ddot{u}_{mx}^{(k)}(0) \right\| \\
 &\leq \frac{1}{2} \left(\left\| u_{mx}^{(k)}(0) \right\| + \left\| F_m^{(k)}(0) \right\| \right)^2 + \frac{1}{2} \left\| \ddot{u}_{mx}^{(k)}(0) \right\|^2 \\
 &\leq \frac{1}{2} \xi_m^{(k)} + \frac{1}{2} \left(\left\| u_{mx}^{(k)}(0) \right\| + \left\| F_m^{(k)}(0) \right\| \right)^2 \\
 &\leq \frac{1}{2} \xi_m^{(k)} + \frac{1}{2} \left(\left\| \tilde{u}_{0kx} \right\| + \left\| \sum_{i+j \leq N-1} D^{ij} f(x, 0, \tilde{u}_0, \tilde{u}_1) (\tilde{u}_{0k} - \tilde{u}_0)^i (\tilde{u}_{1k} - \tilde{u}_1)^j \right\| \right)^2.
 \end{aligned}$$

Therefore,

$$\xi_m^{(k)} \leq \left(\left\| \tilde{u}_{0kx} \right\| + \left\| \sum_{i+j \leq N-1} D^{ij} f(x, 0, \tilde{u}_0, \tilde{u}_1) (\tilde{u}_{0k} - \tilde{u}_0)^i (\tilde{u}_{1k} - \tilde{u}_1)^j \right\| \right)^2. \tag{18}$$

By (9) and (18), we can deduce the existence of a constant $M > 0$ independent of k and m such that

$$\begin{aligned}
 S_m^{(k)}(0) &= \left\| \tilde{u}_{1k} \right\|^2 + \left\| \tilde{u}_{0k} \right\|^2 + 3 \left\| \tilde{u}_{1kx} \right\|^2 + \left\| \tilde{u}_{0kxx} \right\|^2 + \left\| \tilde{u}_{1kxx} \right\|^2 + \xi_m^{(k)} \\
 &\leq \frac{M^2}{4} \quad \text{for all } m, k \in \mathbb{N}.
 \end{aligned} \tag{19}$$

In order to continue other estimates, we establish the following key lemma including properties of $F_m^{(k)}(t)$, $F_{mx}^{(k)}(t)$, and $\dot{F}_m^{(k)}(t)$.

Lemma 3 $F_m^{(k)}(t)$, $F_{mx}^{(k)}(t)$, $\dot{F}_m^{(k)}(t)$ satisfy the following inequalities

$$\left\| F_m^{(k)}(t) \right\|_{L^\infty} \leq \tilde{b}_M \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right], \quad (20)$$

$$\left\| F_{mx}^{(k)}(t) \right\|_{L^\infty} \leq \tilde{b}_M \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right], \quad (21)$$

$$\left\| \dot{F}_m^{(k)}(t) \right\|_{L^\infty} \leq \tilde{b}_M \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right], \quad (22)$$

where

$$\tilde{b}_M = (4M + 2N - 1)(1 + M^{N-1})K_M(f) \sum_{i+j \leq N-1} \frac{1}{i!j!} 2^{i+j}. \quad (23)$$

Proof Applying Lemma 1, using the inequalities $(a+b)^p \leq 2^{p-1}(a^p + b^p)$ for all $a, b > 0$, $p \geq 1$, and

$$s^p \leq 1 + s^{N-1} \quad \forall s \geq 0, \quad \forall p, \quad N, \quad 0 \leq p \leq N-1,$$

we deduce from (5), (13), (17) that

$$\left| u_m^{(k)}(t) - u_{m-1}(t) \right| \leq \left\| u_{mx}^{(k)}(t) \right\| + \left\| \nabla u_{m-1}(t) \right\| \leq \sqrt{S_m^{(k)}(t)} + M; \quad (24a)$$

$$\left| \dot{u}_m^{(k)}(t) - u'_{m-1}(t) \right| \leq \left\| \dot{u}_{mx}^{(k)}(t) \right\| + \left\| \nabla u'_{m-1}(t) \right\| \leq \sqrt{S_m^{(k)}(t)} + M; \quad (24b)$$

$$\begin{aligned} \left(\sqrt{S_m^{(k)}(t)} + M \right)^{i+j} &\leq 2^{i+j-1} \left[\left(\sqrt{S_m^{(k)}(t)} \right)^{i+j} + M^{i+j} \right] \\ &\leq 2^{i+j-1} \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} + 1 + M^{N-1} \right] \\ &\leq 2^{i+j} (1 + M^{N-1}) \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right] \end{aligned} \quad (24c)$$

for all $i, j \in \mathbb{N}$, $1 \leq i+j \leq N-1$. It follows from (10) and (24a)–(24c) that

$$\begin{aligned} \left| F_m^{(k)}(x, t) \right| &\leq K_M(f) + K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} \left| u_m^{(k)}(t) - u_{m-1}(t) \right|^i \left| \dot{u}_m^{(k)}(t) - u'_{m-1}(t) \right|^j \\ &\leq K_M(f) + K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} \left(\sqrt{S_m^{(k)}(t)} + M \right)^{i+j} \\ &\leq K_M(f)(1 + M^{N-1}) \sum_{i+j \leq N-1} \frac{1}{i!j!} 2^{i+j} \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right] \\ &\leq \tilde{b}_M \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right], \end{aligned}$$

where $\tilde{b}_M$ is defined as in (23). Hence, (20) follows.

On the other hand,

$$\begin{aligned}
 F_{mx}^{(k)}(x, t) &= D_1 f[u_{m-1}] + D_3 f[u_{m-1}] \nabla u_{m-1} + D_4 f[u_{m-1}] \nabla u'_{m-1} \\
 &+ \sum_{1 \leq i+j \leq N-1} \left[D_1 D^{ij} f[u_{m-1}] + D_3 D^{ij} f[u_{m-1}] \nabla u_{m-1} + D_4 D^{ij} f[u_{m-1}] \nabla u'_{m-1} \right] \\
 &\quad \times (u_m^{(k)} - u_{m-1})^i (\dot{u}_m^{(k)} - u'_{m-1})^j \\
 &+ \sum_{1 \leq i+j \leq N-1} i D^{ij} f[u_{m-1}] (u_m^{(k)} - u_{m-1})^{i-1} (u_{mx}^{(k)} - \nabla u_{m-1}) (\dot{u}_m^{(k)} - u'_{m-1})^j \\
 &+ \sum_{1 \leq i+j \leq N-1} j D^{ij} f[u_{m-1}] (u_m^{(k)} - u_{m-1})^i (\dot{u}_m^{(k)} - u'_{m-1})^{j-1} (\dot{u}_{mx}^{(k)} - \nabla u'_{m-1}). \quad (25)
 \end{aligned}$$

By Lemma 2, we obtain from (5), (13), (14), and (17) that

$$\begin{aligned}
 |\nabla u_{m-1}(t)| + |\nabla u'_{m-1}(t)| &\leq \|\nabla u_{m-1}(t)\|_{C^0([0,1])} + \|\nabla u'_{m-1}(t)\|_{C^0([0,1])} \\
 &\leq 2 \left(\|\Delta u_{m-1}(t)\| + \|\Delta u'_{m-1}(t)\| \right) \leq 4M; \quad (26a)
 \end{aligned}$$

$$\begin{aligned}
 \left| u_{mx}^{(k)}(t) - \nabla u_{m-1}(t) \right| &\leq 2 \left\| u_{mx}^{(k)}(t) \right\| + 2 \|\Delta u_{m-1}(t)\| \\
 &\leq 2 \left(\sqrt{S_m^{(k)}(t)} + M \right); \quad (26b)
 \end{aligned}$$

$$\begin{aligned}
 \left| \dot{u}_{mx}^{(k)}(t) - \nabla u'_{m-1}(t) \right| &\leq 2 \left\| \dot{u}_{mx}^{(k)}(t) \right\| + 2 \|\Delta u'_{m-1}(t)\| \\
 &\leq 2 \left(\sqrt{S_m^{(k)}(t)} + M \right). \quad (26c)
 \end{aligned}$$

It follows from (24a)–(24c), (25), and (26a)–(26c) that

$$\begin{aligned}
 \left| F_{mx}^{(k)}(x, t) \right| &\leq K_M(f)(1 + |\nabla u_{m-1}(t)| + |\nabla u'_{m-1}(t)|) \\
 &+ K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} (1 + |\nabla u_{m-1}(t)| + |\nabla u'_{m-1}(t)|) \left| u_m^{(k)} - u_{m-1} \right|^i \left| \dot{u}_m^{(k)} - u'_{m-1} \right|^j \\
 &+ K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} i \left| u_m^{(k)} - u_{m-1} \right|^{i-1} \left| u_{mx}^{(k)} - \nabla u_{m-1} \right| \left| \dot{u}_m^{(k)} - u'_{m-1} \right|^j \\
 &+ K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} j \left| u_m^{(k)} - u_{m-1} \right|^i \left| \dot{u}_m^{(k)} - u'_{m-1} \right|^{j-1} \left| \dot{u}_{mx}^{(k)} - \nabla u'_{m-1} \right| \\
 &\leq K_M(f)(1 + 4M) + K_M(f)(1 + 4M) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} \left(\sqrt{S_m^{(k)}(t)} + M \right)^{i+j} \\
 &\quad + K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} 2(i+j) \left(\sqrt{S_m^{(k)}(t)} + M \right)^{i+j} \\
 &\leq K_M(f)(1 + 4M) + K_M(f)(4M + 2N - 1) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} \left(\sqrt{S_m^{(k)}(t)} + M \right)^{i+j} \\
 &\leq K_M(f)(1 + 4M) \\
 &\quad + K_M(f)(4M + 2N - 1)(1 + M^{N-1}) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} 2^{i+j} \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right]
 \end{aligned}$$

$$\begin{aligned} &\leq K_M(f)(4M + 2N - 1)(1 + M^{N-1}) \sum_{i+j \leq N-1} \frac{1}{i!j!} 2^{i+j} \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right] \\ &= \tilde{b}_M \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right]. \end{aligned} \quad (27)$$

Thus, (21) holds, it remains to prove (22). We have

$$\begin{aligned} \dot{F}_m^{(k)}(x, t) &= D_2 f[u_{m-1}] + D_3 f[u_{m-1}]u'_{m-1} + D_4 f[u_{m-1}]u''_{m-1} \\ &+ \sum_{1 \leq i+j \leq N-1} \left[D_2 D^{ij} f[u_{m-1}] + D_3 D^{ij} f[u_{m-1}]u'_{m-1} + D_4 D^{ij} f[u_{m-1}]u''_{m-1} \right] \\ &\quad \times (u_m^{(k)} - u_{m-1})^i (\dot{u}_m^{(k)} - u'_{m-1})^j \\ &+ \sum_{1 \leq i+j \leq N-1} i D^{ij} f[u_{m-1}] (u_m^{(k)} - u_{m-1})^{i-1} (\dot{u}_m^{(k)} - u'_{m-1}) (\dot{u}_m^{(k)} - u'_{m-1})^j \\ &+ \sum_{1 \leq i+j \leq N-1} j D^{ij} f[u_{m-1}] (u_m^{(k)} - u_{m-1})^i (\dot{u}_m^{(k)} - u'_{m-1})^{j-1} (\ddot{u}_m^{(k)} - u''_{m-1}). \end{aligned} \quad (28)$$

By Lemma 1, we deduce from (5), (16), (17) that

$$\begin{aligned} |u'_{m-1}(t)| + |u''_{m-1}(t)| &\leq \|u'_{m-1}(t)\|_{C([0,1])} + \|u''_{m-1}(t)\|_{C([0,1])} \\ &\leq \|\nabla u'_{m-1}(t)\| + \|\nabla u''_{m-1}(t)\| \leq 2M; \end{aligned} \quad (29a)$$

$$\left| \ddot{u}_m^{(k)}(t) - u''_{m-1}(t) \right|^j \leq \left\| \ddot{u}_{mx}^{(k)}(t) \right\| + \|\nabla u''_{m-1}(t)\| \leq \sqrt{S_m^{(k)}(t)} + M. \quad (29b)$$

Hence, we obtain from (24a)–(24c), (28), and (29a)–(29b) that

$$\begin{aligned} \left| \dot{F}_m^{(k)}(x, t) \right| &\leq K_M(f) \left(1 + \|u'_{m-1}(t)\|_{C([0,1])} + \|u''_{m-1}(t)\|_{C([0,1])} \right) \\ &+ K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} \left(1 + \|u'_{m-1}(t)\|_{C([0,1])} + \|u''_{m-1}(t)\|_{C([0,1])} \right) \\ &\quad \times \left| u_m^{(k)} - u_{m-1} \right|^i \left| \dot{u}_m^{(k)} - u'_{m-1} \right|^j \\ &+ K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} i \left| u_m^{(k)} - u_{m-1} \right|^{i-1} \left| \dot{u}_m^{(k)} - u'_{m-1} \right|^{j+1} \\ &+ K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} j \left| u_m^{(k)} - u_{m-1} \right|^i \left| \dot{u}_m^{(k)} - u'_{m-1} \right|^{j-1} \left| \ddot{u}_m^{(k)} - u''_{m-1} \right| \end{aligned}$$

$$\begin{aligned}
 &\leq K_M(f)(1+2M) + K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} (1+2M) \left(\sqrt{S_m^{(k)}(t)} + M \right)^{i+j} \\
 &\quad + K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} (i+j) \left(\sqrt{S_m^{(k)}(t)} + M \right)^{i+j} \\
 &\leq K_M(f)(1+2M) + K_M(f)(2M+N) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} \left(\sqrt{S_m^{(k)}(t)} + M \right)^{i+j} \\
 &\leq K_M(f)(1+2M) + K_M(f)(2M+N)(1+M^{N-1}) \\
 &\quad \times \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} 2^{i+j} \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right] \\
 &\leq K_M(f)(2M+N)(1+M^{N-1}) \sum_{i+j \leq N-1} \frac{1}{i!j!} 2^{i+j} \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right] \\
 &\leq \tilde{b}_M \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right].
 \end{aligned}$$

Therefore, Lemma 3 is proved. $\square$

Now, we estimate all integrals $I_1, \dots, I_3$.

Using Lemma 3, we obtain

$$\begin{aligned}
 I_1 &= 2 \int_0^t \left\langle F_m^{(k)}(s), \dot{u}_m^{(k)}(s) \right\rangle ds \leq 2\tilde{b}_M \int_0^t \left[1 + \left(\sqrt{S_m^{(k)}(s)} \right)^{N-1} \right] \sqrt{S_m^{(k)}(s)} ds \\
 &\leq 2\tilde{b}_M \int_0^t \left[\sqrt{S_m^{(k)}(s)} + \left(\sqrt{S_m^{(k)}(s)} \right)^N \right] ds \\
 &\leq 4\tilde{b}_M \left[T + \int_0^t \left(S_m^{(k)}(s) \right)^N ds \right]. \tag{30}
 \end{aligned}$$

Similarly,

$$I_2 \leq 4\tilde{b}_M \left[T + \int_0^t \left(S_m^{(k)}(s) \right)^N ds \right], \tag{31}$$

$$I_3 \leq 4\tilde{b}_M \left[T + \int_0^t \left(S_m^{(k)}(s) \right)^N ds \right]. \tag{32}$$

Combining (17), (19), (30)–(32), we get

$$S_m^{(k)}(t) \leq \frac{M^2}{4} + 12T\tilde{b}_M + 12\tilde{b}_M \int_0^t \left(S_m^{(k)}(s) \right)^N ds, \quad 0 \leq t \leq T_m^{(k)} \leq T. \tag{33}$$

Solving the nonlinear Volterra integral inequality (33) (based on methods used in [8]), we obtained the following lemma.

Lemma 4 *There exists a constant $T > 0$ independent of k and m such that*

$$S_m^{(k)}(t) \leq M^2 \quad \forall t \in [0, T], \quad \forall k, m \in \mathbb{N}. \tag{34}$$

By Lemma 4, we can take constant $T_m^{(k)} = T$ for all m and k . Therefore,

$$u_m^{(k)} \in W(M, T) \quad \text{for all } m \text{ and } k \in \mathbb{N}. \quad (35)$$

Convergence. By (35), there exists a subsequence of $\{u_m^{(k)}\}$, denoted by the same symbol, such that

$$\begin{cases} u_m^{(k)} \rightarrow u_m & \text{in } L^\infty(0, T; H_0^1 \cap H^2) \text{ weakly}^*, \\ \dot{u}_m^{(k)} \rightarrow u'_m & \text{in } L^\infty(0, T; H_0^1 \cap H^2) \text{ weakly}^*, \\ \ddot{u}_m^{(k)} \rightarrow u''_m & \text{in } L^\infty(0, T; H_0^1) \text{ weakly}^*, \\ u_m \in W(M, T). \end{cases} \quad (36)$$

Applying the compactness lemma of Lions [7, p. 57] and the theorem of Fischer–Riesz, we conclude from (36) that there exists a subsequence of $\{u_m^{(k)}\}$, still denoted by the same symbol, such that

$$\begin{cases} u_m^{(k)} \rightarrow u_m & \text{strongly in } L^2(0, T; H_0^1) \text{ and a.e. in } Q_T, \\ \dot{u}_m^{(k)} \rightarrow u'_m & \text{strongly in } L^2(0, T; H_0^1) \text{ and a.e. in } Q_T. \end{cases} \quad (37)$$

On the other hand, by $L^\infty(0, T; H_0^1 \cap H^2) \hookrightarrow L^\infty(Q_T)$ and by the inequality

$$|a^j - b^j| \leq j M^{j-1} |a - b| \quad \forall a, b \in [-M, M], \quad \forall M > 0, \quad \forall j \in \mathbb{N}, \quad (38)$$

we deduce from (34) that

$$\left| (u_m^{(k)})^j - (u_m)^j \right| \leq j M^{j-1} \left| u_m^{(k)} - u_m \right|, \quad j = 0, 1, \dots, N-1. \quad (39)$$

Therefore, (37) and (39) give

$$\left(u_m^{(k)} \right)^j \longrightarrow (u_m)^j \quad \text{strongly in } L^2(Q_T).$$

We note that

$$\begin{aligned} F_m^{(k)}(x, t) - F_m(x, t) &= D_3 f[u_{m-1}](u_m^{(k)} - u_m) + D_4 f[u_{m-1}](\dot{u}_m^{(k)} - u'_m) \\ &= \sum_{2 \leq i+j \leq N-1} D^{ij} f[u_{m-1}] \left[\left(u_m^{(k)} - u_{m-1} \right)^i - (u_m - u_{m-1})^i \right] \left(\dot{u}_m^{(k)} - u'_{m-1} \right)^j \\ &\quad + \sum_{2 \leq i+j \leq N-1} D^{ij} f[u_{m-1}] (u_m - u_{m-1})^i \left[\left(\dot{u}_m^{(k)} - u'_{m-1} \right)^j - (u'_m - u'_{m-1})^j \right]. \end{aligned}$$

Hence,

$$\begin{aligned}
 & \left| F_m^{(k)}(x, t) - F_m(x, t) \right| \leq K_M(f) \left[\left| u_m^{(k)} - u_m \right| + \left| \dot{u}_m^{(k)} - u'_m \right| \right] \\
 & + K_M(f) \sum_{2 \leq i+j \leq N-1} \frac{1}{i!j!} \left| \left(u_m^{(k)} - u_{m-1} \right)^i - (u_m - u_{m-1})^i \right| \left| \dot{u}_m^{(k)} - u'_{m-1} \right|^j \\
 & + K_M(f) \sum_{2 \leq i+j \leq N-1} \frac{1}{i!j!} |u_m - u_{m-1}|^i \left| \left(\dot{u}_m^{(k)} - u'_{m-1} \right)^j - (u'_m - u'_{m-1})^j \right| \\
 & \leq K_M(f) \left[\left\| u_{mx}^{(k)} - u_{mx} \right\| + \left\| \dot{u}_{mx}^{(k)} - u'_{mx} \right\| \right] \\
 & + K_M(f) \sum_{2 \leq i+j \leq N-1} \frac{1}{i!j!} \left| \left(u_m^{(k)} - u_{m-1} \right)^i - (u_m - u_{m-1})^i \right| \left\| \dot{u}_{mx}^{(k)} - \nabla u'_{m-1} \right\|^j \\
 & + K_M(f) \sum_{2 \leq i+j \leq N-1} \frac{1}{i!j!} (2M)^i \left| \left(\dot{u}_m^{(k)} - u'_{m-1} \right)^j - (u'_m - u'_{m-1})^j \right|. \quad (40)
 \end{aligned}$$

On the other hand, we deduce from (34) and (38) that

$$\begin{aligned}
 \left| \left(u_m^{(k)} - u_{m-1} \right)^i - (u_m - u_{m-1})^i \right| & \leq i(2M)^{i-1} \left| u_m^{(k)} - u_m \right| \\
 & \leq i(2M)^{i-1} \left\| u_{mx}^{(k)} - u_{mx} \right\|, \quad (41)
 \end{aligned}$$

$$\begin{aligned}
 \left| \left(\dot{u}_m^{(k)} - u'_{m-1} \right)^j - (u'_m - u'_{m-1})^j \right| & \leq j(2M)^{j-1} \left| \dot{u}_m^{(k)} - u'_m \right| \\
 & \leq j(2M)^{j-1} \left\| \dot{u}_{mx}^{(k)} - u'_{mx} \right\|. \quad (42)
 \end{aligned}$$

It follows from (40)–(42) that

$$\begin{aligned}
 & \left| F_m^{(k)}(x, t) - F_m(x, t) \right| \leq K_M(f) \left[\left\| u_{mx}^{(k)} - u_{mx} \right\| + \left\| \dot{u}_{mx}^{(k)} - u'_{mx} \right\| \right] \\
 & + K_M(f) \sum_{2 \leq i+j \leq N-1} \frac{1}{i!j!} i(2M)^{i-1} \left\| u_{mx}^{(k)} - u_{mx} \right\| (2M)^j \\
 & + K_M(f) \sum_{2 \leq i+j \leq N-1} \frac{1}{i!j!} (2M)^i j(2M)^{j-1} \left\| \dot{u}_{mx}^{(k)} - u'_{mx} \right\| \\
 & \leq K_M(f) \left[\left\| u_{mx}^{(k)} - u_{mx} \right\| + \left\| \dot{u}_{mx}^{(k)} - u'_{mx} \right\| \right] \\
 & + NK_M(f) \sum_{2 \leq i+j \leq N-1} \frac{1}{i!j!} (2M)^{i+j-1} \left\| u_{mx}^{(k)} - u_{mx} \right\| \\
 & + NK_M(f) \sum_{2 \leq i+j \leq N-1} \frac{1}{i!j!} (2M)^{i+j-1} \left\| \dot{u}_{mx}^{(k)} - u'_{mx} \right\| \\
 & \leq NK_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} (2M)^{i+j-1} \left[\left\| u_{mx}^{(k)} - u_{mx} \right\| + \left\| \dot{u}_{mx}^{(k)} - u'_{mx} \right\| \right] \\
 & = \tilde{d}_M \left[\left\| u_{mx}^{(k)} - u_{mx} \right\| + \left\| \dot{u}_{mx}^{(k)} - u'_{mx} \right\| \right],
 \end{aligned}$$

where

$$\tilde{d}_M = N K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} (2M)^{i+j-1}.$$

Thus

$$\left\| F_m^{(k)} - F_m \right\|_{L^2(Q_T)} \leq \tilde{d}_M \left[\left\| u_{mx}^{(k)} - u_{mx} \right\|_{L^2(Q_T)} + \left\| \dot{u}_{mx}^{(k)} - \dot{u}_{mx}' \right\|_{L^2(Q_T)} \right] \rightarrow 0,$$

so (37) leads to

$$F_m^{(k)} \rightarrow F_m \quad \text{strongly in } L^2(Q_T).$$

Passing to limit in (8), (9), we have u_m satisfying (6), (7) in $L^2(0, T)$.

On the other hand, it follows from (6)₁ and (36)₄ that

$$\frac{\partial^2}{\partial x^2} (u_m''(t) + u_m(t)) = u_m''(t) - F_m(t) \in L^\infty(0, T; H_0^1).$$

Consequently,

$$u_m''(t) + u_m(t) = \Phi \in L^\infty(0, T; H_0^1 \cap H^2),$$

so

$$u_m''(t) = \Phi - u_m(t) \in L^\infty(0, T; H_0^1 \cap H^2).$$

Hence, $u_m \in W_1(M, T)$ and Theorem 1 is proved. $\square$

3 The Convergence at a Rate of Order N

We put

$$W_1(T) = \{v \in L^\infty(0, T; H_0^1) : v' \in L^\infty(0, T; H_0^1)\},$$

then $W_1(T)$ is a Banach space with respect to the norm

$$\|v\|_{W_1(T)} = \|v\|_{L^\infty(0, T; H_0^1)} + \|v'\|_{L^\infty(0, T; H_0^1)}.$$

Theorem 2 Suppose that $\tilde{u}_0, \tilde{u}_1 \in H_0^1 \cap H^2$ and $f \in C^N([0, 1] \times \mathbb{R}_+ \times \mathbb{R}^2)$ satisfying $f(0, t, 0, 0) = f(1, t, 0, 0) = 0 \forall t \geq 0$. Then, there exist constants $M > 0$ and $T > 0$ such that

- (i) Problem (1)–(3) has a unique weak solution $u \in W_1(M, T)$.
- (ii) The sequence $\{u_m\}$, defined as in Theorem 1, converges at a rate of order N to the solution u strongly in the space $W_1(T)$ in the sense

$$\|u_m - u\|_{W_1(T)} \leq C \|u_{m-1} - u\|_{W_1(T)}^N, \quad (43)$$

for all $m \geq 1$, where C is a suitable constant. On the other hand, the estimate is fulfilled

$$\|u_m - u\|_{W_1(T)} \leq C_T \beta^{N^m} \quad \text{for all } m \in \mathbb{N}, \quad (44)$$

where C_T and $0 < \beta < 1$ are the constants depending only on T .

Proof Existence. Put $w_m = u_{m+1} - u_m$, it is obvious that w_m satisfies the variational problem

$$\begin{cases} \langle w_m''(t), w \rangle + \langle w_{mx}(t) + w_{mx}'(t), w_x \rangle = \langle F_{m+1}(t) - F_m(t), w \rangle & \forall w \in H_0^1, \\ w_m(0) = w_m'(0) = 0. \end{cases} \quad (45)$$

Taking $w = w'_m$ in (45), after integrating in t , we get

$$Z_m(t) = 2 \int_0^t \langle F_{m+1}(s) - F_m(s), w'_m(s) \rangle ds, \quad (46)$$

where

$$Z_m(t) = \|w'_m(t)\|^2 + \|w_{mx}(t)\|^2 + \|w'_{mx}(t)\|^2. \quad (47)$$

Using Taylor's expansion for the function $f[u_m] = f(x, t, u_m, u'_m) = f(x, t, u_{m-1} + w_{m-1}, u'_{m-1} + w'_{m-1})$ around the point u_{m-1} up to order N , we obtain

$$\begin{aligned} f[u_m] &= f[u_{m-1}] + \sum_{1 \leq i+j \leq N-1} D^{ij} f[u_{m-1}] w_{m-1}^i (w'_{m-1})^j \\ &\quad + \sum_{i+j=N} D^{ij} f[\tilde{u}_{m-1}] w_{m-1}^i (w'_{m-1})^j, \end{aligned} \quad (48)$$

where $[\tilde{u}_{m-1}] = (x, t, u_{m-1} + \theta_1 w_{m-1}, u'_{m-1} + \theta_1 w'_{m-1})$, $0 < \theta_1 < 1$.

Hence, it follows from (7) and (48) that

$$\begin{aligned} F_{m+1}(x, t) - F_m(x, t) &= f[u_m] - f[u_{m-1}] + \sum_{1 \leq i+j \leq N-1} D^{ij} f[u_m] w_m^i (w'_m)^j \\ &\quad - \sum_{1 \leq i+j \leq N-1} D^{ij} f[u_{m-1}] w_{m-1}^i (w'_{m-1})^j \\ &= \sum_{1 \leq i+j \leq N-1} D^{ij} f[u_m] w_m^i (w'_m)^j \\ &\quad + \sum_{i+j=N} D^{ij} f[\tilde{u}_{m-1}] w_{m-1}^i (w'_{m-1})^j. \end{aligned}$$

Therefore,

$$\begin{aligned} \|F_{m+1}(t) - F_m(t)\| &\leq K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} \|w_{mx}(t)\|^i \|w'_{mx}(t)\|^j \\ &\quad + K_M(f) \sum_{i+j=N} \frac{1}{i!j!} \|\nabla w_{m-1}(t)\|^i \|\nabla w'_{m-1}(t)\|^j \\ &\leq K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} \left(\sqrt{Z_m(t)} \right)^{i+j} + K_M(f) \sum_{i+j=N} \frac{1}{i!j!} \|w_{m-1}\|_{W_1(T)}^{i+j} \\ &\leq K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} M^{i+j-1} \sqrt{Z_m(t)} + K_M(f) \sum_{i+j=N} \frac{1}{i!j!} \|w_{m-1}\|_{W_1(T)}^N \\ &\leq \gamma_T \sqrt{Z_m(t)} + \tilde{\gamma}_T \|w_{m-1}\|_{W_1(T)}^N, \end{aligned} \quad (49)$$

where

$$\gamma_T = K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} M^{i+j-1}, \quad \tilde{\gamma}_T = K_M(f) \sum_{i+j=N} \frac{1}{i!j!}.$$

Then we deduce from (46), (47), and (49) that

$$\begin{aligned} Z_m(t) &\leq 2 \int_0^t \|F_{m+1}(s) - F_m(s)\| \|w'_m(s)\| ds \\ &\leq 2 \int_0^t \left[\gamma_T^{(1)} \sqrt{Z_m(s)} + \tilde{\gamma}_T \|w_{m-1}\|_{W_1(T)}^N \right] \sqrt{Z_m(s)} ds \\ &\leq 2\gamma_T \int_0^t Z_m(s) ds + 2\tilde{\gamma}_T \int_0^t \|w_{m-1}\|_{W_1(T)}^N \sqrt{Z_m(s)} ds \\ &\leq T \tilde{\gamma}_T \|w_{m-1}\|_{W_1(T)}^{2N} + (2\gamma_T + \tilde{\gamma}_T) \int_0^t Z_m(s) ds. \end{aligned} \quad (50)$$

By using Gronwall's lemma, (50) leads to

$$\|w_m\|_{W_1(T)} \leq \mu_T \|w_{m-1}\|_{W_1(T)}^N, \quad (51)$$

where $\mu_T = 2\sqrt{\tilde{\gamma}_T T \exp((2\gamma_T + \tilde{\gamma}_T)T)}$. Then, it follows from (51) that, for all m and p ,

$$\|u_m - u_{m+p}\|_{W_1(T)} \leq (1 - \beta)^{-1} (\mu_T)^{\frac{1}{N-1}} \beta^{Nm}. \quad (52)$$

Choose T small enough such that $\beta = M\mu_T^{\frac{1}{N-1}} < 1$. It follows that $\{u_m\}$ is a Cauchy sequence in $W_1(T)$. Then there exists $u \in W_1(T)$ such that

$$u_m \longrightarrow u \quad \text{strongly in } W_1(T). \quad (53)$$

Note that $u_m \in W_1(M, T)$, then there exists a subsequence $\{u_{m_j}\}$ of $\{u_m\}$ such that

$$\begin{cases} u_{m_j} \rightarrow u & \text{in } L^\infty(0, T; H_0^1 \cap H^2) \text{ weakly}^*, \\ u'_{m_j} \rightarrow u' & \text{in } L^\infty(0, T; H_0^1 \cap H^2) \text{ weakly}^*, \\ u''_{m_j} \rightarrow u'' & \text{in } L^\infty(0, T; H_0^1) \text{ weakly}^*, \\ u \in W(M, T). \end{cases}$$

We have

$$\begin{aligned} |F_m(x, t) - f[u]| &\leq |f[u_{m-1}] - f[u]| + K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} |u_m - u_{m-1}|^i |u'_m - u'_{m-1}|^j \\ &\leq K_M(f) [|u_{m-1} - u| + |u'_{m-1} - u'|] \\ &\quad + K_M(f) \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} |u_m - u_{m-1}|^i |u'_m - u'_{m-1}|^j \\ &\leq K_M(f) [\|u_{m-1} - u\|_{W_1(T)} + \sum_{1 \leq i+j \leq N-1} \frac{1}{i!j!} \|u_m - u_{m-1}\|_{W_1(T)}^{i+j}]. \end{aligned} \quad (54)$$

Then, it implies from (53) and (54) that

$$F_m(t) \rightarrow f[u] \quad \text{strongly in } L^\infty(Q_T).$$

Finally, passing to limit in (6), (7) as $m = m_j \rightarrow \infty$, we find $u \in W(M, T)$ satisfying the equation

$$\langle u''(t), w \rangle + \langle u_x(t) + u_x''(t), w_x \rangle = \langle f[u], w \rangle$$

for all $w \in H_0^1$ and the initial conditions

$$u(0) = \tilde{u}_0, \quad u'(0) = \tilde{u}_1.$$

Uniqueness. Applying a similar argument used in the proof of Theorem 1, we conclude that $u \in W_1(M, T)$ is unique.

Passing to the limit in (52) as $p \rightarrow +\infty$ for fixed m , we get (44). In the same way, (43) follows. Theorem 2 is proved completely. $\square$

Remark 1 In order to construct an N -order iterative scheme, we need the condition $f \in C^N([0, 1] \times \mathbb{R}_+ \times \mathbb{R}^2)$. Then, we obtain a convergent sequence at a rate of order N to a local weak solution of the problem. This condition of f can be relaxed if we only consider the existence of solution, see [9–12, 15, 16].

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